

Chapter 10

Problems 1

Starting from this chapter we will begin solving problems by capitalizing precisely on the theoretical machinery of inversion of the real plane with respect to a circle that we have developed so far. At first, we will be solving carefully orchestrated problems that are practically designed to be solved specifically using inversion of the real plane in a circle.

Later on, as we gain more confidence, we will be solving problems that admit a number of distinctly different types of solutions and it is not at all obvious that the machinery of inversion of the real plane in a circle may have anything to do with such problems. However, upon a closer inspection, we will see that with some of such problems inversion of the real plane in a circle will lead to victory.

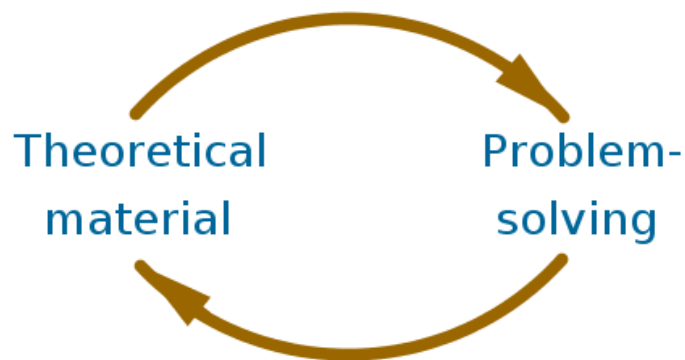
Why

Having covered some basic amount of theoretical territory, we now arrive at a juncture that is very important in any vaguely defined layer of mathematics, solving technical problems.

The activity of solving technical problems in any mathematical domain has two main purposes.

The first purpose of solving technical problems in any mathematical domain is to integrate the possibly dry, detached and abstract theoretical material into a practical context where the students, while expanding their web of technical knowledge, get to feel and experience with their own skin what the theory is all about and what the tactical might of the developed theory has to offer, can do and can achieve.

During the solution of the domain-specific technical problems the back-propagation process actually delivers the miracle of the improved understanding of the relevant theoretical material, which, in turn, enables the students to take on deeper and more difficult problems, which, through the said back-propagation channel, improve the understanding of the theoretical material, which, in turn ... (Figure 10.1):



The second purpose of solving technical problems in any mathematical domain is to develop and improve the students' problem-solving skills and tactical prowess.

Different people have different opinions about the technical problem-solving for the sake of developing the art and the craft of technical problem-solving. Some people like to participate in Olympic, competitive, events and have a blast doing it, while some other people think that this is mostly a waste of time and that a student of mathematics should get on with the serious business of moving the mathematical needle.

Whatever your personal stance on technical problem-solving is and before you even develop such a stance and jump to a conclusion, do keep in mind that solving a large number of properly designed technical problems is one of the secrets of your success.

Technical problems have at least two degrees of freedom.

The first degree of freedom of technical problems is their texture, which varies wildly from the routine plug-and-chug silly homework problems to the problems that are actually of research level in their own right. Such research level problems knock on the door of Big Science.

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Our advice to the readers is to not waste their life on the former and concentrate on the latter.

The second degree of freedom of technical problems is their objective, of which we distinguish two types:

- a problem to find (or a problem to compute, or a problem to construct) and
- a problem to prove

As an example, compare the following two problems.

A problem to find. Find the value of the following integral:

$$\int_0^{+\infty} \frac{x^a}{1+x^3} \cdot \frac{dx}{x}$$

A problem to prove. Under the assumption that f is a continuous function and that the following integral converges absolutely:

$$\int_0^{+\infty} f(x) \cdot x^a \cdot \frac{dx}{x}$$

prove that that integral is equal to:

$$\int_{-\infty}^{+\infty} f(e^t) \cdot e^{at} dt$$

As such, in this part of the book we present a tiny smidgen of technical problems all of which do require the students to engage into a flavor of the activity that is often referred to as “thinking mathematically” or “thinking like a mathematician”.

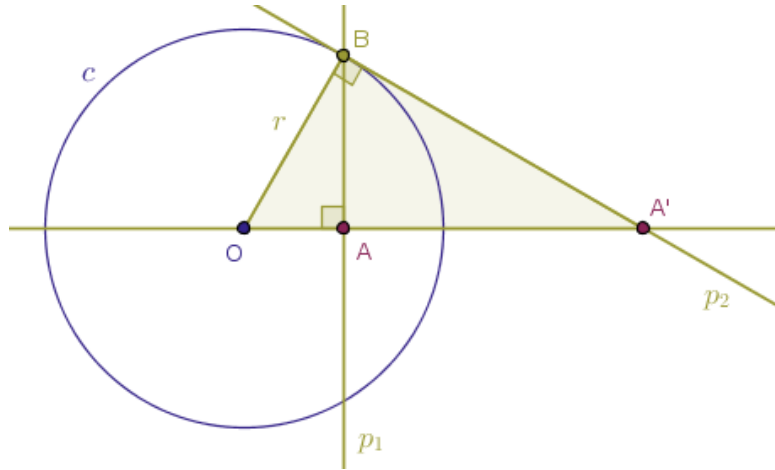
In the upcoming challenges there will be no plug-and-chug problems whatsoever.

Every problem presented here will require the reader to think independently on her/his own. Every such problem is a nano dress rehearsal for the bigger journey ahead. Every such problem will require the reader to reason from first principles, analyze the situation and craft a mathematically sound argument that takes the problem-solver to victory.

It goes without saying that the reader is expected to digest the relevant problem statement only, walk away from the computer screen and attempt to solve the current problem on her/his own for only in that case the above side effects of problem-solving will have a chance to manifest themselves.

As a trivial opener, we observe in passing that due to the results of our Problem 1 and Problem 3, it is always possible to use the inversion with respect to a unit circle to construct a reciprocal of a given length on the cheap.

For example, let $g = |OA|$ be the length of a given line segment, any which one whatsoever. Then, clearly, as per the drawing of Figure 3.5:



the length of the line segment OA' will be the sought-after reciprocal of the length g if, that is, we set the length r of the radius of the circle of inversion c to the positive unity:

$$|OA'| = \frac{r}{|OA|} = \frac{1}{|OA|} = \frac{1}{g}, \quad r \equiv 1$$

and conversely. If g is given by $|AO'|$ then the length $|OA|$ will be the reciprocal of the length $g = |OA'|$.

On a more serious note, we have identified the following inversion-based problem-solving approach.

A Description

We pedantically use the word *approach* because whatever we are about to describe next is most certainly not an algorithm and not even a method. Both an algorithm and a method imply that if followed correctly, they will produce an answer sought-after that is correct.

With our approach, however, there are no guarantees of any kind. That is. In order to make an attempt to solve a given technical problem using the machinery of inversive geometry, we should try to:

- choose a fruitful center of inversion
- choose a fruitful radius of inversion
- choose a fruitful power of inversion or a combination thereof

- invert the object(s) given with respect to a circle found in the previous steps a fruitful number of times
- solve a presumably easier problem in terms of the resultant inverse object(s)
- using the fundamental properties of inversive geometry, such as the preservation of angles and tangency points, convert the solution in terms of the inverse object(s) into the sought-after solution in terms of the original object(s)

Note that our suggestion to “choose a fruitful whatever” is hardly algorithmic. Which is the point. As mathematically inclined thinkers, we develop the nose and the gut for a successful choice of relevant objects only with time and only by putting our own boots on the ground and solving a large number of domain-specific problems.

We also note in passing that the above approach is a part of a larger and a more generic, universally applicable, problem-solving pattern or a heuristic template known as “Transform And Conquer” and that can be bluntly described thus:

- transform
- solve
- reverse transform

For the sake of brevity and in order to avoid the tiresome and burdensome repetition, we take it that all the objects of interest in this book are always coplanar with the given or input objects, unless noted otherwise.

With all the technicalities out of the way, let us now study in great detail exactly how the above approach may be used in practice.

Tangent Circles

To that end we will solve a sample problem that can be solved trivially, without any effort, as is and without the application of the machinery of inversive geometry. However, by taking on a problem that is as simple as possible but not simpler, we will be able to concentrate exactly on the mechanics of the application of the said machinery of inversive geometry without getting distracted on irrelevant details.

Problem 11. Construct all circles that are tangent to a given circle, $q(Q, R)$, pass through a given fixed (coplanar) point, P , and have their centers located on the straight line passing through the points P and Q .

Solution. Based on the mutual juxtaposition of the given fixed point, P , and the given circle, q , the two interesting cases in this problem arise when the point P is located strictly outside of the circle q and when that point is located strictly inside of the circle q .

While we will discuss the solution of this problem as it pertains to the first case, our readers are encouraged to address the second case on their own.

Draw the straight line of centers of the circles sought-after, l , that passes through the points P and Q until that straight line intersects the given circle, q , at two distinct and diametrically opposite points, A and B .

Taking the suggested list of step from the top, we now address the

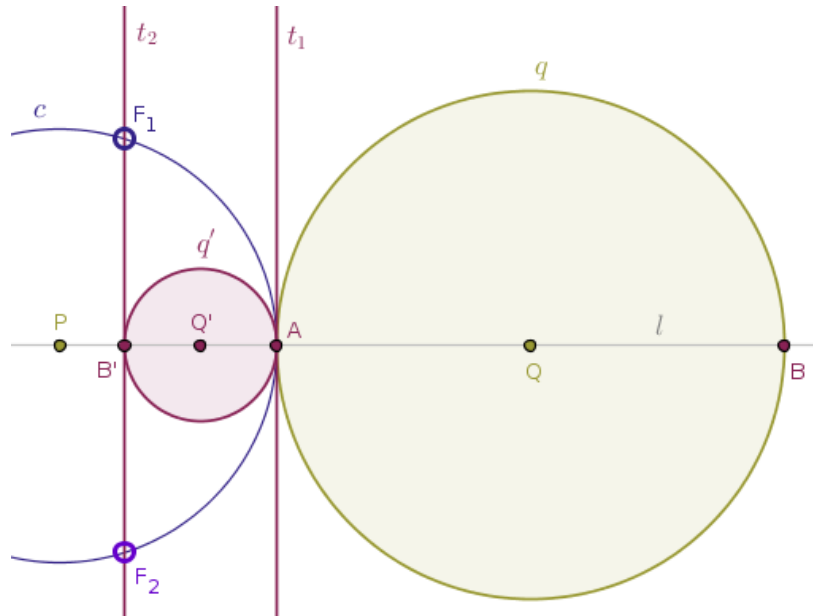
Center of Inversion

We know that the circles sought-after must pass through the given point P . We also remember that under the inversion of the real plane with respect to a circle with both powers the circles that pass through the center of inversion become straight lines which we deem as easier to construct and conversely.

Under the inversion of the real plane with respect to a circle with both powers certain straight lines become circles that pass through the center of inversion. Thus, in this problem it makes sense to pick the given point P as the center of inversion because if we somehow construct the said straight lines and invert them back in a circle that has the point P as its center then we should obtain our circles sought-after.

Radius of Inversion

More often than not, the actual magnitude of the radius of inversion is irrelevant (Figure 10.2):



That is why the length of the radius of inversion can be safely set to the positive unity in many cases. In general, then, the decision about the actual value of the radius of inversion is driven by the following optimization criterion:

whenever we can, we would like to minimize the number of objects that we have to invert

Hence, in this particular problem it makes sense to choose the length of the line segment PA as the radius of inversion but the reader is encouraged to experiment with choosing the length of the line segment PB as the radius of inversion instead: $r = |PA|$.

Power of Inversion

More often than not, we will consider the inversion of the ambient plane with respect to a circle with positive power by default. However, there do exist problems where the inversion with respect to a circle with negative power is the winning ticket, see the Problem 19, An Arbelos Encounter, in the Chapter 14 of this book.

In this particular problem we will investigate the inversion with respect to a circle with positive power.

Invert

Now we invert the tactically advantageous objects using the parameters of inversion that we have chosen in the previous steps. That is, we invert the ambient plane in the circle $c(P, |PA|)$ with positive power.

We know that under the inversion of the real plane with respect to the circle $c(P, |PA|)$ with positive power:

- the point B will be thrown inside of the circle of inversion, c , as the point B'
- the point A will remain fixed and
- the given circle $q(Q, R)$ will become a circle, $q'(Q', R' = |Q'A| = |Q'B'|)$, that touches the circle of inversion, c , at the point A internally

Note well that from both the theoretical and the practical material that we have developed so far it follows that all the steps listed above can be carried out with Euclidean tools alone and according to the well-established algorithms all of which have been described prior.

Solution in Terms of Inverse Objects

Here we observe that this is place where we win because the solution of the problem that is equivalent to the given one in terms of the inverse object is a pair of straight lines, t_1 and t_2 , each of which is tangent to the image circle $q'(Q', R')$ at the points A and B' correspondingly and at this point the construction of a tangent to a circle through a point on that circle is not only “a previously solved problem” but a problem that is inherently trivial.

Since a given circle possesses a unique tangent straight line that passes through a given point on the said circle, it follows that as long as the points A and B' remain distinct there will be two and only two distinct solutions of this problem in terms of the inverse objects.

One more time, we would like to emphasize the fact that we consider drawing **straight lines** tangent to a circle as an easier and a well-known and a well-algorithmisized task or a previously solved problem to which we managed to reduce the current problem, which constitutes our tactical win.

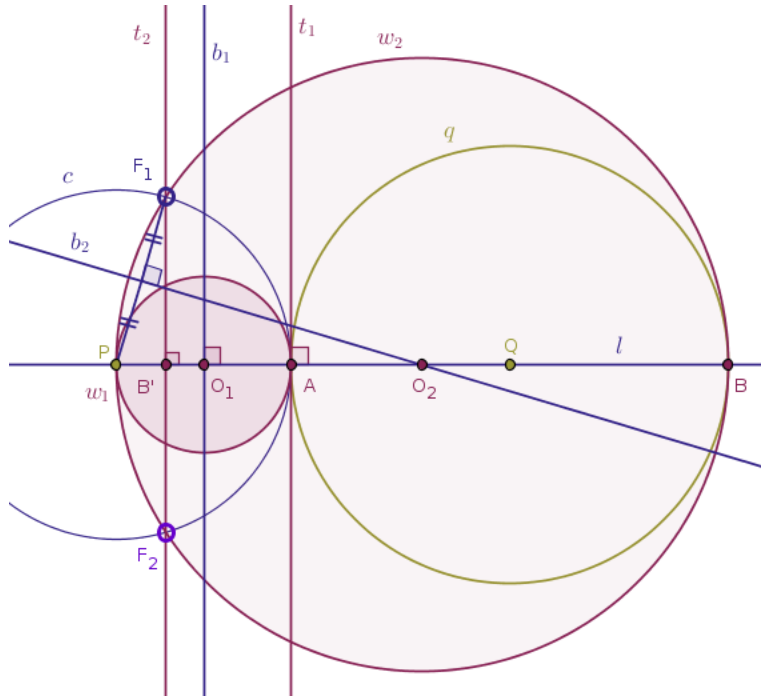
It may also help to say the following words out loud: we now obtained two objects that are tangent to the inverse image of a given object.

Conversion of the Solution in Terms of Inverse Objects

Under the chosen inversion, by the Theorem 12, the straight line t_1 will be inverted into a circle, w_1 , that:

- passes through the given point P
- touches the circle of inversion, c , at the point A **internally**
- touches the given circle, q , at the point A **externally**

But we know that the tangency points under any inversion of the real plane with respect to a circle with both powers are preserved by the Theorem 37. Hence, if the inverted images t_1 and q' touch (at the image point A) then the corresponding original objects w_1 and q will touch also (at the original, fixed, point A) and this is exactly what we want (Figure 10.3):



One down, one to go: invert the straight line t_1 with respect to the circle $c(P, |PA|)$ with positive power once. Moreover, we even know how to locate the center, O_1 , of the circle w_1 : bisect the line segment AP with a straight line, b_1 , until that straight line intersects the straight line of centers, l , at the sought-after point O_1 . Thus, we have constructed the first solution circle, $w_1(O_1, |AP|/2)$.

Moving on. Our straight line t_2 intersects the circle of inversion, c , at the points F_1 and F_2 . Under the chosen inversion, by the Theorem 10, that straight line t_2 will be inverted into a circle, w_2 , that:

- passes through the given point P
- intersects the circle of inversion, c , at the points F_1 and F_2 and
- touches the given circle, q , at the point B internally

But we know that the tangency points under the inversion with respect to a circle with both powers are preserved. Hence, if the inverted images t_2 and q' touch (at the image point B') then the corresponding original objects, w_2 and q , will touch also (at the original point B') and this is exactly what we want.

Two down, zero to go: invert the straight line t_2 with respect to the circle $c(P, |PA|)$ with positive power once. Moreover, we even know how to quickly locate the center, O_2 , of the circle w_2 : bisect either the line segment PF_1 or PF_2 with a straight line, b_2 , until that straight line intersects the line of centers, l , at the sought-after point, Q_2 . Finally, because in general we have been solving this problem in reverse order, we now officially record

The Solution In The Sought-after Forward-flow-of-causality Order

- draw a straight line, l , through the given points P and Q until that straight line intersects the given circle, q , at the points A and B
- invert the given circle, q , with respect to the circle $c(P, |PA|)$ with positive power, once, into its image circle, q'
- obtain the point, B' , where the straight line l and the image circle q' intersect
- draw a straight line, t_1 , tangent to the image circle q' at the point A (a win)
- draw a straight line, t_2 , tangent to the image circle q' at the point B (another win)
- invert the straight line t_1 with respect to the circle $c(P, |PA|)$ with positive power into a circle, w_1 (the first solution)
- invert the straight line t_2 with respect to the circle $c(P, |PA|)$ with positive power into a circle w_2 (the second solution)

We, thus, see that this particular problem admits two distinct solutions. $\square$

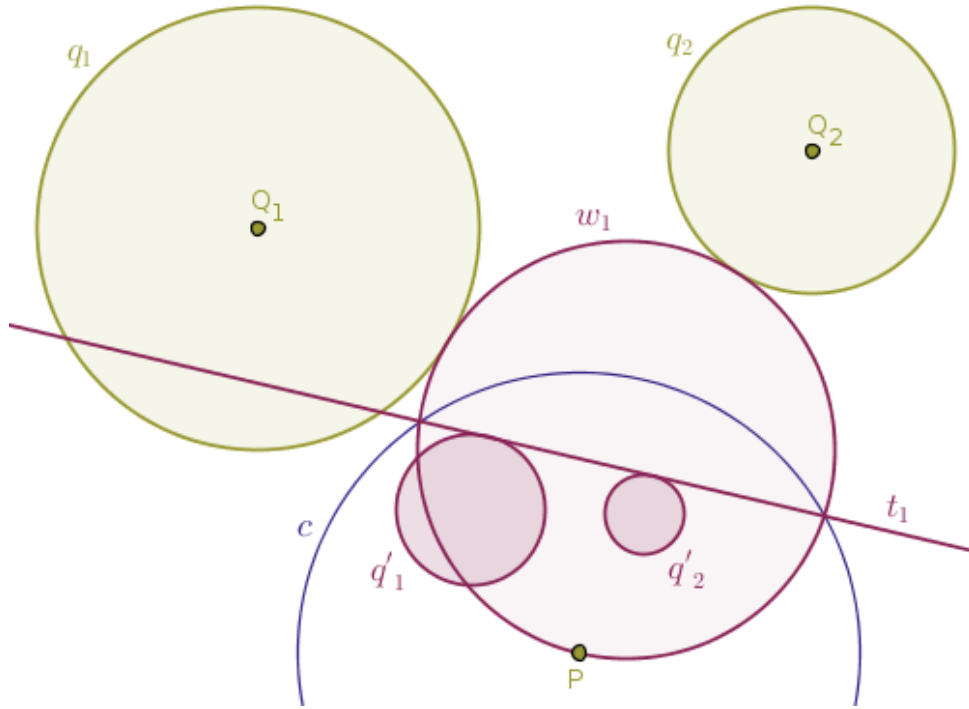
Now that we have covered one sample solution in great detail, we shall move at a faster pace.

Problem 12. Construct all circles that are tangent to two given circles, $q_1(Q_1, R_1)$ and $q_2(Q_2, R_2)$, and pass through a given point, P , located outside of the given circles, q_1 and q_2 .

Solution. As usual, we attempt to solve this, constructive, problem in reverse order by, first, assuming that the object(s) sought-after have been constructed somehow and, then, by trying to unwind the resultant configuration into a traditional or forward-flow-of-causality solution.

Thus, let us assume for a moment that one of the sought-after circles, name that circle w_1 , has been magically constructed already. That circle passes, as required, through the given point P . Hence, we already know that choosing the given point P as the center of inversion with positive power is a good idea.

We now quickly enumerate the inverted objects. Under the inversion of the ambient plane with respect to the circle $c(P, r)$, where the specific length of the radius of inversion, r , in this case is mostly irrelevant, the two given green circles q_1 and q_2 , by the Theorem 30 for example, will be taken into the deep purple circles q'_1 and q'_2 , while the sought-after circle w_1 , by the Theorem 15, will be taken into the straight line t_1 (Figure 10.4):



We also already know that since the original object w_1 touches the given circles q_1 and q_2 , the inverse image of w_1 , which is a straight line t_1 , will also touch the inverse images q'_1 and q'_2 . In other words, under the chosen inversion the straight line t_1 will play the role of a straight line that is tangent to two circles.

But the above observation solves this problem completely.

Why?

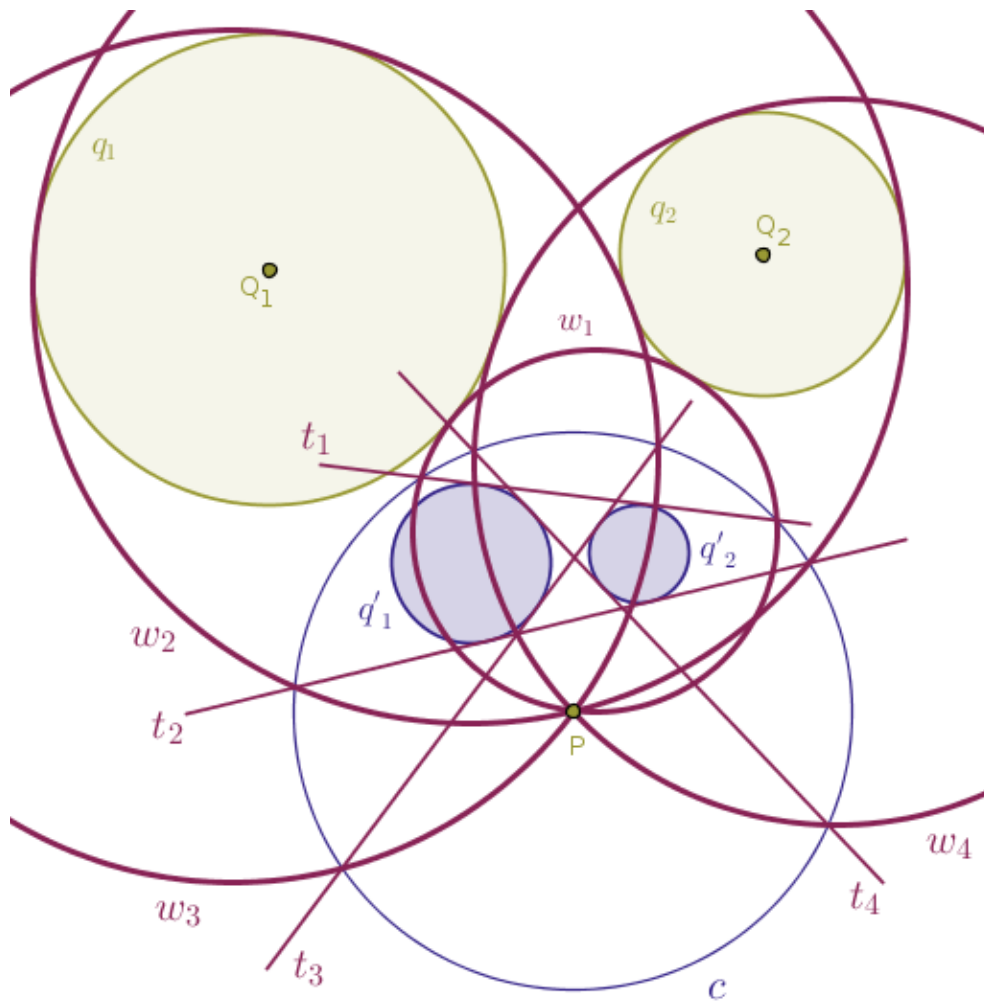
Because the task of drawing a **straight line** (t_1) that is tangent to two circles (q'_1 and q'_2) is a previously solved problem and a (much) simpler problem at that. This is where we win tactically.

In the most general case, how many distinct common tangent straight lines can be drawn to two distinct circles?

Four. Two **internal** and two **external** straight tangent lines. Thus, the overall forward-flow-of-causality solution should be obvious by now.

Step 1. Invert the given circles, q_1 and q_2 , with respect to a circle $c(P, r)$ with positive power to obtain the circles q'_1 and q'_2 .

Step 2. Draw the four distinct common straight lines t_{1-4} tangent to the image circles q'_1 and q'_2 (Figure 10.5):



Step 3. Invert the four distinct straight lines t_{1-4} with respect to the circle $c(P, r)$ with positive power into four distinct circles, w_{1-4} , sought-after.

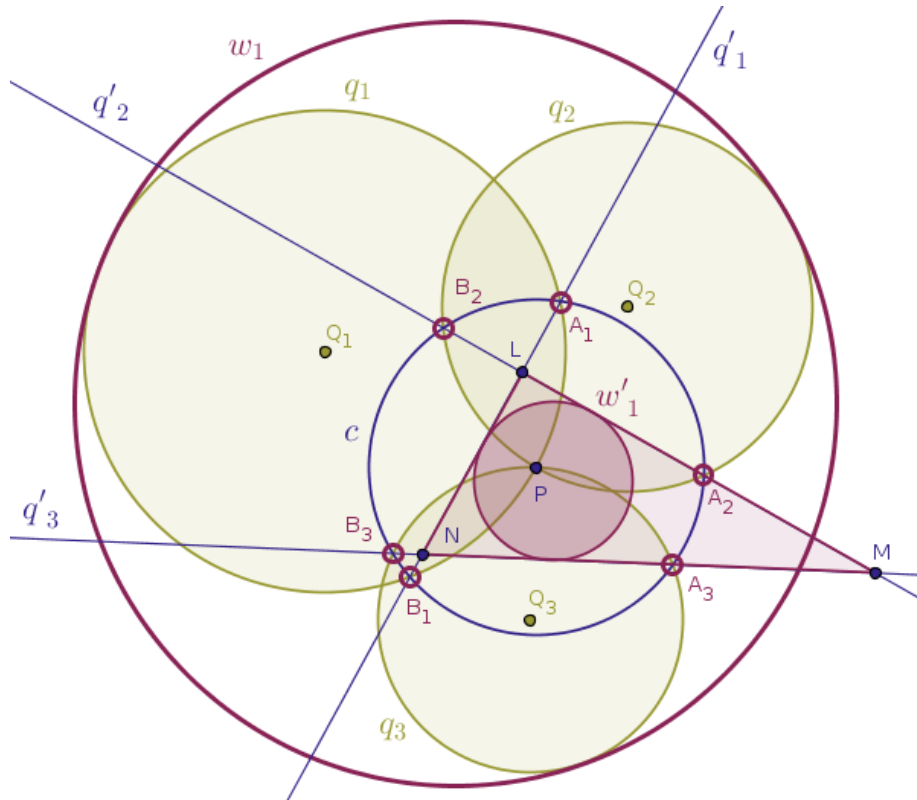
In the most general case the four distinct solution circles will have the following properties:

- one solution circle (w_1) will touch both given circles **externally**
- one solution circle (w_2) will touch both given circles **internally**
- one solution circle (w_3) will touch one of the given circles (q_1) **internally**, while touching the other given circle (q_2) **externally** and vice versa (w_4)

In the drawing above we show all the solutions circles w_{1-4} in deep purple with heavier lines and we pair up the straight lines t_k with the corresponding solution circles w_k based on the subscript k : to the straight line t_1 there corresponds the solution circle w_1 and so on. $\square$

Problem 13. Given three distinct circles, q_1 , q_2 and q_3 , which intersect at a point, P , construct all circles that are tangent to these three circles, q_1 , q_2 and q_3 .

Solution. Choose the given point, P , to be the center of inversion. Choose the radius of inversion, r , in such a way that the circle of inversion $c(P, r)$ intersects all three given distinct circles (Figure 10.6):



Then, under the inversion of the parent plane with respect to such a circle $c(P, r)$ with positive power all three given circles q_1, q_2 and q_3 will, by the Theorem 15, be taken into three distinct **straight lines** $q'_{1,2,3}$.

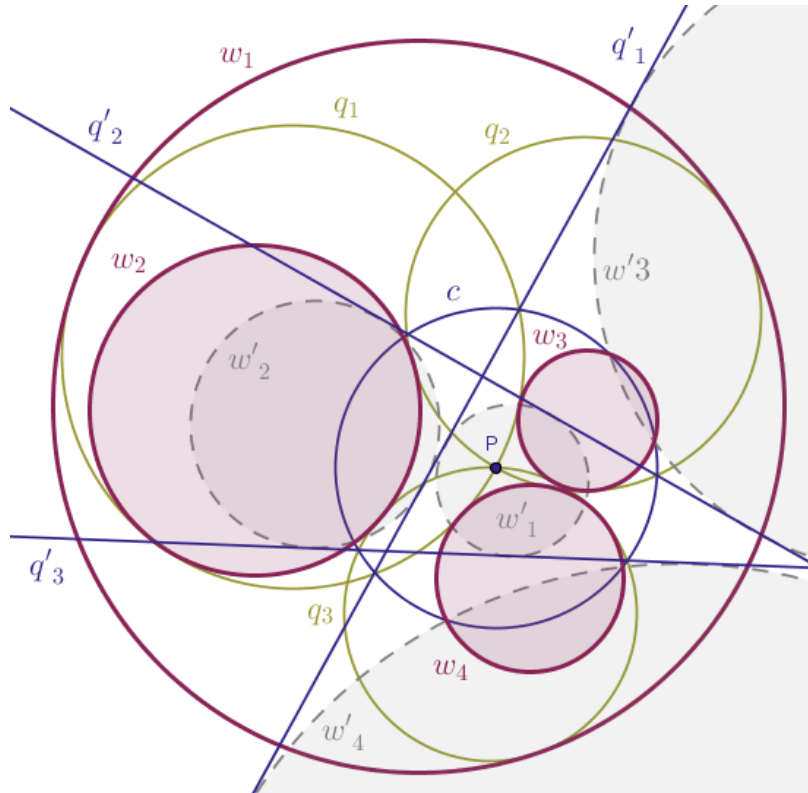
But because the original circles pairwise intersect, but not touch, it follows that their images, the straight lines $q'_{1,2,3}$, will also pairwise intersect, thus, forming a triangle, LMN .

Such a triangle has four distinct circles each of which is tangent to all three straight lines forming the said triangle, one **incircle** and three **excircles**.

Therefore, if we invert these four distinct circles with respect to the circle $c(P, r)$ with positive power once then, due to the fact that the tangency points between these circles and the straight lines $q'_{1,2,3}$ will be preserved, we will obtain the solution circles sought-after.

In the diagram above we show a sample **incircle** w'_1 of the triangle LMN and its image under the inversion with respect to the circle $c(P, r)$ with positive power, the circle w_1 that touches all three given circles **internally**. Thus, the overall forward-flow-of-causality solution should be obvious by now:

Step 1. Draw a circle, $c(P, r)$, that intersects all three given circles (Figure 10.7):



Step 2. Invert the given circles q_1, q_2 and q_3 with respect to the circle $c(P, r)$ with positive power once in order to obtain the straight lines q'_1, q'_2 and q'_3 that will form a triangle, τ .

Step 3. Construct that triangle τ 's single unique **incircle**, w'_1 , and three unique and distinct **excircles**, $w'_{2,3,4}$, all of which will touch the straight lines $q'_{1,2,3}$.

Step 4. Invert the circles w'_1, w'_2, w'_3 and w'_4 with respect to the circle $c(P, r)$ with positive power, once, in order to obtain the four distinct circles, $w_{1,2,3,4}$, sought after.

In the Figure 10.7 above we show the incircle w'_1 and the excircles $w'_{2,3,4}$ in solid gray color with dashed lines. We also pair up the circles $w'_{1,2,3,4}$ with their inverse images $w_{1,2,3,4}$ based on the subscript: for example, under the chosen inversion the circle w'_1 is transformed into the circle w_1 , the circle w'_2 is transformed into the circle w_2 and so on. $\square$